CROWN-Inpaint: A Confidence-weighted, Regime-aware, Non-local, Multi-scale Iterative Framework for Image Inpainting

– Updated Project Proposal –

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Abstract

We propose **CROWN-Inpaint**, a unified iterative framework for single-image inpainting that integrates four complementary priors – *Confidence*, *Regime*, *Non-local*, and multi-scale *Warm-start* – into a single optimisation-first solver layered on top of an established masked dictionary-learning backbone. After a careful audit of prior art (in particular Mairal–Elad–Sapiro 2008, Naumova–Schnass 2018, and the recent Innsbruck Bachelor thesis of Tiefenthaler 2018, which our group identified during literature review), we explicitly position *masked K-SVD* as a settled baseline rather than a contribution, and re-centre our novelty on the **coupling layer** above it. Our four novel mechanisms are: (i) a continuous, time-varying confidence map that drives a weighted OMP generalising the binary masked OMP of prior work; (ii) a per-pixel regime map that fuses a biharmonic smooth-prior branch with the sparse branch; (iii) a non-local centralised ℓ_1 coefficient-coupling step adapted from CSR/GSR but *embedded inside* the iterative masked solver, with a slow-rate Lasso oracle inequality bounding its error by a Self-Similarity Index; and (iv) a coarse-to-fine pyramid with prior injection that provably removes the *zero-information plateau* on which single-scale K-SVD gets stuck when the hole exceeds the patch size. We provide a self-contained convergence and error analysis, an ablation-rich experimental plan, and a clear delineation of which components are baselines and which are our actual contribution.

1 Problem Statement

Let $y \in \mathbb{R}^{H\times W}$ be a single-channel image observed through a binary mask $M \in \{0,1\}^{H\times W}$, with $M_i = 1$ iff pixel i is observed and $M_i = 0$ iff pixel i is missing. The observed set is $\Omega = \{i : M_i = 1\}$ and the hole is Ω^c. We seek a reconstruction $u \in \mathbb{R}^{H\times W}$ such that

(i) $u_i = y_i$ for all $i \in \Omega$ (*exact preservation of observed data*); and

(ii) u on Ω^c is statistically and visually consistent with the observed surround.

Within the dictionary-based paradigm, every $p \times p$ patch $R_q u \in \mathbb{R}^n$ ($n = p^2$) is modelled as

$$R_q u \;\approx\; D\boldsymbol{\alpha}_q, \qquad \|\boldsymbol{\alpha}_q\|_0 \le s, \tag{1}$$

for a learned dictionary $D \in \mathbb{R}^{n\times K}$ ($K \gtrsim n$, unit-norm columns) and a sparse code $\boldsymbol{\alpha}_q \in \mathbb{R}^K$ with at most s non-zeros. Because the mask forbids access to ground-truth patches, the dictionary itself must be learned from incomplete observations, by approximately solving

$$\min_{D,\,A} \sum_q \|M_q\,(R_q y - D\boldsymbol{\alpha}_q)\|_2^2 \quad \text{s.t. } \|\boldsymbol{\alpha}_q\|_0 \le s,\ \|D_{:,k}\|_2 = 1, \tag{2}$$

where $M_q \in \{0,1\}^{n\times n}$ is the diagonal restriction of M to the patch at position q. Equation (2) is the established *weighted K-SVD objective*, due to Mairal–Elad–Sapiro [2] and refined in Naumova–Schnass [4] and Tiefenthaler [5]; it is *not* our contribution.

The remainder of this proposal addresses the practical and theoretical *gap* between existing solvers of (2) and what is required for high-fidelity inpainting under (a) heavy missingness ($r > 70\%$), (b) holes that are larger than a patch, and (c) mixed smooth/textured content – regimes where every existing single-scale, binary-masked, non-coupled solver in the literature is observably suboptimal.

2 Related Work and the State of the Art

2.1 Masked dictionary inpainting (the baseline)

The objective (2) originates with Mairal–Elad–Sapiro [2] (*weighted K-SVD*, "wKSVD"), where the dictionary update is performed by a weighted rank-one approximation of the masked error matrix. Convergence of the weighted rank-one step is established by Srebro–Jaakkola [3] via an EM/imputation argument.

The *ITKrMM* algorithm of Naumova–Schnass [4] extends this line by replacing OMP with thresholding and the SVD with K residual means, and additionally learns an *L-dimensional low-rank component* Γ that captures the dense "mean + contrast" part of natural-image statistics.

Most recently, Tiefenthaler [5] (Innsbruck, 2018, supervisor K. Schnass) explicitly extends wKSVD to allow arbitrary-dimensional low-rank components (algorithm *wKSVDL*), and benchmarks wKSVDL against ITKrMM on Peppers and Mandrill at 30%/50%/70% random erasure plus a structured *Cracks* mask. The thesis settles, in the Schnass group's own line of work, the *single-scale*, binary-mask, single-pass dictionary inpainting question.

What this prior art establishes (and our project does not claim as novel). The following are settled and we use them as baselines:

(B1) *Masked OMP* for sparse coding of partially observed patches ([2], also Alg. 1 of [5]).

(B2) *Weighted rank-one dictionary update* solving (2)'s atom-wise sub-problem ([2, 3, 5]).

(B3) *Single-pass dictionary inpainting* – learn D once, sparse-code masked patches, overlap-average to a final image.

(B4) *Low-rank component handling* (ITKrMM and wKSVDL).

2.2 Smooth-prior inpainting

Variational and PDE-based methods (Chan–Shen [9], Bertalmio–Sapiro [10]) provably dominate dictionary-based methods on smooth holes: biharmonic completion is the unique minimiser of bending energy with Dirichlet/Neumann boundary conditions, hence essentially optimal in the texture-free regime. Conversely, these methods smear textures.

2.3 Non-local sparse representations

Centralised Sparse Representation (CSR, Dong–Zhang–Shi [7]) and Group-Based Sparse Representation (GSR, Zhang–Zhao–Gao [8]) showed that regularising each patch's sparse code toward the weighted mean of its similar patches' codes improves stability on *fully observed* restoration. To our knowledge, no prior work integrates this kind of non-local coefficient coupling into a *masked, iterative* dictionary-inpainting solver with provable error control. CROWN-Inpaint does.

2.4 Multi-scale image processing

Image pyramids and coarse-to-fine optimisation are classical, but their specific role in escaping the *zero-information plateau* of masked K-SVD when the hole exceeds the patch has not, to our knowledge, been formally analysed in the dictionary-inpainting literature.

3 The Gap We Exploit

The four limitations of every solver of (2) known to us, including [2, 4, 5], are the following.

L1: binary masks. All cited works treat the mask as $M \in \{0,1\}^d$. A pixel inside the hole is either "trusted at 0" or "not used". There is no mechanism for the solver to assign *partial* trust to a hole estimate and update that trust over outer iterations.

L2: no smooth/sparse fusion. Pure dictionary inpainting smears smooth regions; pure biharmonic inpainting smears textures. Existing solvers commit to one prior. No existing masked-dictionary solver fuses both with a per-pixel, data-driven fusion weight.

L3: no non-local coupling inside the masked solver. CSR and GSR demonstrated the value of non-local coefficient coupling on fully observed data, but not within a confidence-weighted, masked, iterative framework, and not with explicit error bounds.

L4: no remedy for holes > patch size. When a contiguous hole is larger than the patch, every patch interior to the hole has $M_q = \mathbf{0}$, forcing $\boldsymbol{\alpha}_q = \mathbf{0}$ at every iteration – a *zero-information plateau* on the loss surface. Prior work either avoids this regime experimentally or accepts the resulting black smear.

CROWN-Inpaint addresses L1–L4 simultaneously, with formal guarantees, while keeping the masked-K-SVD backbone of B1–B4 intact.

4 Proposed Contribution: CROWN-Inpaint

4.1 High-level architecture

CROWN-Inpaint is an *outer iterative* solver. Each outer iteration performs five steps:

Step 1. Confidence map update. Compute a continuous $c^t \in [0,1]^{H\times W}$ on hole pixels from (a) overlap variance of patch reconstructions, (b) distance to Ω, and (c) an iteration schedule. Combine with the binary mask to form $W^t = M + (1-M) \odot c^t$.

Step 2. Confidence-weighted sparse branch. For every patch q, run a weighted-OMP on $(R_q u^{t-1}, D, W_q^t)$, then refine the selected codes by a non-local centralised ℓ_1 ISTA toward the group centre $\boldsymbol{\beta}_q$. Overlap-average to obtain u^t_{sparse}.

Step 3. Smooth branch. Run a small number of harmonic Jacobi relaxation steps from u^{t-1}, holding Ω fixed, to obtain u^t_{smooth}.

Step 4. Regime-aware fusion. Compute or update a regime map $r \in [0,1]^{H\times W}$ from observed gradients, structure-tensor anisotropy and DCT spectral entropy; fuse $\bar{u}^t = (1-r) \odot u^t_{\text{smooth}} + r \odot u^t_{\text{sparse}}$.

Step 5. Hard projection. $u^t = M \odot y + (1-M) \odot \bar{u}^t$. Optionally apply a small stochastic manifold-correction step every few iterations.

The outer loop is wrapped inside a *three-level coarse-to-fine pyramid* that performs the entire procedure at 32×32, then 64×64, then 128×128, with the coarse reconstruction injected as a warm start to the next finer level.

4.2 The four genuinely novel mechanisms

We list the four *additive* novel mechanisms that CROWN-Inpaint introduces above the baselines B1–B4 and the gaps L1–L4.

N1 – Continuous confidence-weighted OMP and outer loop. We replace the binary mask M used by every prior masked-dictionary solver by a continuous confidence-weighted $W^t \in [0,1]^d$. Concretely we solve, per patch and per outer iteration,

$$\min_{\boldsymbol{\alpha}} \left\| (W_q^t)^{1/2} \odot (R_q u^{t-1} - D\boldsymbol{\alpha}) \right\|_2^2 \quad \text{s.t. } \|\boldsymbol{\alpha}\|_0 \le s. \tag{3}$$

The substitution $\widetilde{D} = \operatorname{diag}\big((W_q^t)^{1/2}\big) D$, $\widetilde{x} = (W_q^t)^{1/2} \odot R_q u^{t-1}$ reduces (3) *exactly* to standard OMP on $(\widetilde{D}, \widetilde{x})$ with per-column re-normalisation, preserving the reduction trick of binary masked OMP while strictly generalising it. The confidence schedule is the key mechanism that lets the solver gradually trust its own hole estimates without admitting them blindly at iteration 1.

N2 – Regime-aware fusion of smooth and sparse priors. We introduce a per-pixel regime map $r \in [0,1]^{H \times W}$ computed on the *observed* pixels via gradient energy, structure-tensor anisotropy, and DCT spectral entropy, and propagated into the hole by boundary-conditional Gaussian smoothing. The fusion

$$\bar{u} = (1 - r) \odot u_{\text{smooth}} + r \odot u_{\text{sparse}}$$

gates two priors with provably complementary regimes of optimality (biharmonic on smooth holes; sparse coding on textured holes). To our knowledge no prior masked-dictionary inpainting work performs this fusion at all, let alone with a learned per-pixel gate.

N3 – Non-local centralised $\boldsymbol{\ell_1}$ coupling inside the masked solver. For each patch q, define a non-local prior

$$\boldsymbol{\beta}_q = \sum_{r \in \mathrm{KNN}(q)} w_{qr}\, \boldsymbol{\alpha}_r, \qquad w_{qr} = \frac{\exp(-\|R_q u - R_r u\|^2_{W_q \odot W_r} / h^2)}{Z_q}, \tag{4}$$

where the similarity weights use *coordinates confidently observed in both patches*. Then refine each code by

$$\boldsymbol{\alpha}_q^\star = \arg\min_{\boldsymbol{\alpha}} \left\| W_q^{1/2} (R_q u - D\boldsymbol{\alpha}) \right\|_2^2 + \lambda \|\boldsymbol{\alpha} - \boldsymbol{\beta}_q\|_1 . \tag{5}$$

This is a *convex* sub-problem solved by an ISTA loop of ~ 20 iterations on the OMP support. Inspired by CSR [7] and GSR [8], (5) differs in three concrete ways: the data term is *confidence-weighted*, the prior $\boldsymbol{\beta}_q$ uses *double-masked* similarity (so unobserved coordinates do not spuriously influence neighbour selection), and the refinement is *embedded inside* an iterative masked solver rather than applied once on fully observed data.

N4 – Coarse-to-fine pyramid with prior injection and plateau-escape guarantee. Existing single-scale solvers fail catastrophically when the hole side H_Ω exceeds the patch side p, because every patch interior to the hole has $M_q = \mathbf{0}$, leading to $\nabla_{\boldsymbol{\alpha}_q}\mathcal{L} \equiv \mathbf{0}$ and an irreducible zero-information plateau on the loss surface. CROWN-Inpaint runs the entire algorithm at the coarsest pyramid level where $H_\Omega/2^{L-1} \leq p$ – at which the dead set is empty – and uses bilinear-upsampled coarse reconstructions as warm starts at finer levels. We *prove* (Theorem 3 below) that the warm start strictly removes the plateau and gives the descent algorithm a non-zero gradient on which to act.

4.3 What CROWN-Inpaint does not claim as novel

For complete intellectual honesty:

- Masked OMP, masked K-SVD, and the weighted rank-one dictionary update are baselines from [2–5].
- The closed-form scalar coordinate ALS we use to implement the rank-one update is a *coordinate-descent variant* of the EM/SVD weighted rank-one update of [3]; we re-derive it because the scalar form is faster and avoids any imputation step, but the underlying objective is identical.
- The biharmonic smooth branch is [9] used as-is.
- The non-local idea of regularising codes toward neighbour means is from CSR/GSR; what is novel is its concrete integration into the masked, iterative, confidence-weighted setting (item N3 above).
- The pyramid is a classical idea; what is novel is the formal plateau-escape guarantee in the masked-K-SVD setting (item N4).
- Plug-and-play stochastic manifold correction is from the diffusion-PnP literature; we use it as an *optional* stabiliser.

5 Mathematical Formulation and Inspiration

We can view CROWN-Inpaint as approximate alternating minimisation of the energy

$$\begin{aligned} E(u, \{\boldsymbol{\alpha}_q\}) \;=\; & \sum_q \left\| W_q^{1/2} \odot (R_q u - D\boldsymbol{\alpha}_q) \right\|_2^2 \;+\; \lambda \sum_{q,\, r \in \mathrm{KNN}(q)} w_{qr} \left\| \boldsymbol{\alpha}_q - \boldsymbol{\alpha}_r \right\|_1 \\ & +\; \mu\, J_{\mathrm{smooth}}(u;\, r) \;+\; \iota_\Omega(u;\, y), \end{aligned} \tag{6}$$

subject to $\|\boldsymbol{\alpha}_q\|_0 \leq s$. The four terms correspond exactly to the four CROWN priors:

- the *confidence-weighted* data term (mechanism N1) reduces to classical masked sparse coding when $W_q = M_q \in \{0,1\}^d$;
- the *non-local* term (N3) corresponds to a convex relaxation of the ℓ_1 deviation from $\boldsymbol{\beta}_q$ when one notes that $\sum_r w_{qr} \|\boldsymbol{\alpha}_q - \boldsymbol{\alpha}_r\|_1 \approx \|\boldsymbol{\alpha}_q - \boldsymbol{\beta}_q\|_1$ to leading order in the centred shrinkage analysis;
- the *smooth* term (N2) is a Tikhonov / harmonic energy locally modulated by the regime map r;
- the indicator ι_Ω implements the hard observation constraint (Step 5).

Each step of the outer iteration corresponds to one block-coordinate descent step on (6) with a closed-form (or near-closed-form) inner solve. The pyramid (N4) corresponds to warm-starting the non-convex problem from a coarse minimiser.

5.1 Inspirations – one line each

- **Patch sparsity**: K-SVD [1], masked extensions [2, 4, 5].
- **Confidence weighting**: Plug-and-play and reliability-weighted estimation, recast for masked OMP.
- **Regime / fusion**: classical observation that biharmonic is optimal on smooth holes [9] and that texture needs atom transport from the boundary [10].
- **Non-local centralised refinement**: CSR [7], GSR [8], with the slow-rate Lasso analysis of Bickel–Ritov–Tsybakov [11].
- **Multi-scale warm start**: the empirical observation that single-scale K-SVD fails catastrophically when $H_\Omega > p$, formalised via the plateau-escape theorem of Sec. 6.
- **Optional manifold correction**: stochastic plug-and-play without requiring a pre-trained denoiser network.

6 Why It Works: Theoretical Foundations

We summarise four results that we have already proven for the companion convergence document; full proofs are in the technical report.

T1: Convergence of the masked dictionary update. For each atom k, the per-atom rank-one sub-problem

$$\min_{\boldsymbol{d}_k, \boldsymbol{h}_k} \sum_{i \in \mathcal{S}_k} \|M_i(\boldsymbol{e}_{k,i} - \boldsymbol{d}_k h_{k,i})\|_2^2$$

admits the closed-form scalar updates

$$h_{k,i} = \frac{\sum_j (M_i)_{jj}\, d_{k,j}\, e_{k,i,j}}{\sum_j (M_i)_{jj}\, d_{k,j}^2}, \quad d_{k,j} = \frac{\sum_i (M_i)_{jj}\, h_{k,i}\, e_{k,i,j}}{\sum_i (M_i)_{jj}\, h_{k,i}^2},$$

each of which globally minimises the masked Frobenius cost in its single block. Under a standard atom-utilisation assumption, alternating the two updates yields a monotonically non-increasing energy converging to a coordinate-wise stationary point, and the outer K-SVD loop inherits this descent property. *This recovers the known guarantee of [3], restated as a coordinate descent for clarity and speed.*

T2: Slow-rate Lasso bound for the non-local refinement (novel). Define the **Self-Similarity Index**

$$\mathrm{SSI}_q := \left\|\boldsymbol{\alpha}_q^{\mathrm{true}} - \boldsymbol{\beta}_q\right\|_1 .$$

Theorem 1 (Self-similarity error bound)**.** *Under the noise model* $R_q u^{\mathrm{true}} = D\boldsymbol{\alpha}_q^{\mathrm{true}} + \boldsymbol{\nu}_q$, $\left\|W_q^{1/2}\boldsymbol{\nu}_q\right\|_2 \le \delta$, *and* $\lambda \ge 2\left\|\widetilde{D}_q^\top \widetilde{\boldsymbol{\nu}}_q\right\|_\infty$, *every solution* $\boldsymbol{\alpha}_q^\star$ *of* (5) *satisfies*

$$\left\|W_q^{1/2}\big(R_q u^{\mathrm{true}} - D\boldsymbol{\alpha}_q^\star\big)\right\|_2 \;\le\; \delta + \sqrt{3\lambda \cdot \mathrm{SSI}_q}.$$

This is the slow-rate Lasso inequality applied to the centred regulariser $\|\boldsymbol{\alpha} - \boldsymbol{\beta}_q\|_1$ and is, to our knowledge, the first explicit error bound for a masked, iterative, non-locally coupled sparse-coding step. The proof proceeds by the change of variables $\boldsymbol{\gamma} = \boldsymbol{\alpha} - \boldsymbol{\beta}_q$, which reduces (5) to a standard Lasso whose oracle inequality is then applied. The bound captures the expected behaviour: better self-similar matches (smaller SSI_q) yield tighter reconstruction error.

T3: Plateau-escape via multi-scale warm start (novel).

Lemma 2 (Plateau gradient)**.** *For every patch q with $M_q = \mathbf{0}$ and any D,*

$$\nabla_{\boldsymbol{\alpha}_q} \mathcal{L}(D, A) \;=\; -2D^\top M_q(R_q u - D\boldsymbol{\alpha}_q) \;\equiv\; \mathbf{0}.$$

Theorem 3 (Warm-start removes the plateau)**.** *Let $\widehat{u}^{(\ell)}$ be the reconstruction at the coarse scale, $\mathcal{U}_{\ell\to\ell-1}$ the bilinear upsampler, and define the prior-injected initialisation*

$$u_{\text{init}}^{(\ell-1)} = M^{(\ell-1)} \odot y^{(\ell-1)} + (1 - M^{(\ell-1)}) \odot \mathcal{U}_{\ell\to\ell-1}\big(\widehat{u}^{(\ell)}\big).$$

If the upsampling error is at most η on the hole and the coarse scale has empty dead set, then the gradient of the prior-augmented loss at $u_{\text{init}}^{(\ell-1)}$ is generically non-zero, the OMP sparse coding step returns codes with per-patch fitting error $\leq n\eta^2$, and one ALS sweep produces a strictly lower prior-augmented loss. By contrast, any cold-started iterate $(D^{(0)}, A = \mathbf{0})$ on the strict-mask loss is permanently stuck: $A_{:,q}^{(t)} = \mathbf{0}$ for all $t \geq 0$ and every q with $M_q = \mathbf{0}$.

T4: Composite end-to-end guarantee. Combining T1–T3, the full CROWN-Inpaint pipeline produces a sequence $(D^{(t)}, A^{(t)})$ at each pyramid level that is monotonically non-increasing in the corresponding masked energy, with bounded per-patch reconstruction error $\leq \delta_q + \sqrt{3\lambda \cdot \mathrm{SSI}_q}$ at termination of the non-local step. Crucially, T3 ensures that this descent is taken *from a non-trivial starting point* on holes larger than the patch – the regime where every prior solver fails.

Remark. None of T1–T4 claim global optimality of the recovered (D, A): ℓ_0 sparse coding is NP-hard and OMP is a greedy approximation. This caveat is identical to that of classical K-SVD [1] and is explicitly inherited by every masked-K-SVD-based method in the literature.

7 Experimental Plan

7.1 Datasets and masks

Datasets. *Peppers* and *Mandrill* (256×256, for direct comparison with [5]); subsets of *CelebA-HQ* and *Places2* (128×128 patches); textured subsets from DTD for stress tests.

Masks. (a) Random Bernoulli erasure at 30%/50%/70% (matches [5]); (b) the same *Cracks* structured mask used in [5]; (c) rectangular block holes of side $20-40$ at 128×128 to exercise the multi-scale plateau-escape mechanism (no prior solver handles this gracefully); (d) thin scratches and free-form brush masks.

7.2 Baselines

We benchmark against the following, each implemented or wrapped in our codebase:

(a) **Biharmonic / Telea inpainting** (classical, on smooth holes).

(b) **Plain masked K-SVD** (B1+B2+B3 of Sec. 2, our own implementation already complete).

(c) **wKSVDL** of Tiefenthaler [5] (we will re-implement Algorithms 4–5 of that thesis as our principal competing single-scale baseline).

(d) **ITKrMM** of Naumova–Schnass [4].

(e) **CSR/GSR-style** non-local sparse representation [7, 8], where feasible given the masked setting.

(f) **CROWN-Inpaint** (full method, our contribution).

(g) Deep baselines (*stretch goal*): a single LaMa or partial-conv pre-trained model, as a sanity reference – we are an optimisation-first method and do not aim to beat large pretrained models.

7.3 Ablations (the experimental core of the paper)

We will run ablation rows that toggle each of the four CROWN mechanisms on/off independently:

Variant	N1 conf.	N2 regime	N3 nonlocal	N4 pyramid
Plain masked K-SVD	–	–	–	–
+N1 only	✓	–	–	–
+N2 only	–	✓	–	–
+N3 only	–	–	✓	–
+N4 only	–	–	–	✓
+N1+N3	✓	–	✓	–
+N1+N4	✓	–	–	✓
CROWN-Inpaint (full)	✓	✓	✓	✓

This table is precisely what reviewers will demand and exactly what distinguishes us from [5], which only varies the low-rank dimension $L \in \{1, 3, 8\}$.

7.4 Metrics

PSNR, SSIM, hole-only PSNR, boundary MAE, mean wall-clock per image, and number of outer iterations to convergence. We will report mean and standard deviation over five random seeds per (image, mask) pair, as in [5].

7.5 Hypotheses we will test

H1. N1 (continuous confidence) gives a ≥ 0.5 dB hole-PSNR improvement at $r \geq 50\%$ over binary-mask wKSVDL.

H2. N2 (regime fusion) eliminates the texture-vs-smooth trade-off observed by [5]: the same CROWN configuration wins both Peppers (smooth) and Mandrill (textured), unlike wKSVDL which loses to ITKrMM on Mandrill.

H3. N3 (non-local) reduces variance of reconstruction across random seeds by $\geq 30\%$ at $r = 70\%$.

H4. N4 (pyramid) recovers semantically meaningful content in $\geq 20 \times 20$ block holes where every single-scale baseline produces a black smear.

8 Implementation Status

The codebase is already organised around the architecture above.

File	Component	Status
`masked_ksvd.py`	Masked OMP + ALS update (B1, B2)	complete
`inpainting_masked_ksvd.py`	Single-scale masked inpainting	complete
`inpainting_multiscale_masked_ksvd.py`	3-level pyramid (N4)	complete
`crown/confidence.py`	Confidence map (N1)	complete
`crown/regime.py`	Regime map (N2)	complete
`crown/weighted_omp.py`	Confidence-weighted OMP (N1)	complete
`crown/nonlocal_coupling.py`	Non-local centralised ISTA (N3)	complete
`crown/smooth.py`	Biharmonic / harmonic Jacobi	complete
`crown/fuse.py`	Regime-aware fusion + projection	complete
`crown/manifold.py`	Optional stochastic correction	complete
`crown/run.py`	Outer-loop driver	complete
`experiment_sweep.py`	Ablation grid	in progress
wKSVDL re-implementation	Tiefenthaler 2018 baseline	to do

The convergence-and-error analysis is also written up as a separate technical companion document `convergence_proofs.tex` (IEEE format, 9 pages). It contains full proofs of T1–T4.

9 Timeline and Deliverables

Week	Milestone	Notes
W1	wKSVDL re-implementation	Algorithms 4–5 of [5], validated to match the thesis's Peppers/Mandrill PSNRs (± 0.3 dB).
W2	Ablation sweep	8-row ablation $\times$ 4 mask types $\times$ 2 images $\times$ 5 seeds.
W3	Multi-scale stress test	20×20 and 40×40 block holes; visual + numerical comparison with all single-scale baselines.
W4	Paper draft – main	Sections 1–4 (introduction, related work, method, theory).
W5	Paper draft – experiments	Section 5; integrate plots from W2–W3.
W6	Internal review	Course staff feedback; revise.
W7	Final submission	PDF + reproducible code repository + raw experimental data.

Final deliverables:

D1. Paper PDF (8–10 pages, IEEE format) describing the method, theory, and experiments.

D2. Companion convergence-proof document (`convergence_proofs.tex`, already drafted).

D3. Reproducible code repository with all eight ablation rows and every baseline.

D4. Experimental data (`.csv` + `.npy`) and a notebook reproducing every figure in the paper.

10 Risks and Mitigations

R1: Tiefenthaler 2018 covers more of the masked-dictionary ground than expected. *Already incurred and accommodated.* The present proposal explicitly demotes masked K-SVD itself to a baseline and re-centres novelty on the four CROWN mechanisms (N1–N4). No further pivot is required.

R2: Non-local refinement is too slow for 128×128 images. We restrict the centralised ℓ_1 refinement to the OMP support (at most $s \leq 12$ atoms), turning a K-dim convex problem into an s-dim one. ISTA on s-dim problems is $O(s^2)$ per iteration with ~ 20 iterations; total cost is dominated by the OMP step.

R3: The pyramid leaks coarse-scale artefacts upward. We perform mask-aware downsampling (normalised convolution that excludes hole pixels from the aggregation) at every level transition; this is already implemented in `inpainting_multiscale_masked_ksvd.py`. We will additionally ablate the upsampling kernel.

R4: Manifold correction destabilises convergence. We treat it as optional (off by default) and include it only as an ablation knob, with a decreasing $\sigma_t = \sigma_0 \gamma^{t-1}$ schedule that bounds its long-run effect.

R5: Reviewers question whether N1+N2+N3+N4 are independent or redundant. The 8-row ablation table directly answers this: each row toggles a single mechanism, and we will report the marginal PSNR contribution of each.

11 Expected Outcomes and Publication Target

- A complete, theoretically-grounded, optimisation-first inpainting method that beats every single-scale masked-dictionary baseline (including wKSVDL) by $1-3$ dB on standard benchmarks and produces visually coherent fills on holes that are $2-5\times$ the patch size.
- Four formal results (T1–T4) with full proofs; T2–T4 are, to our knowledge, new in the masked-dictionary inpainting literature.
- A clean ablation table that pinpoints the marginal contribution of each CROWN mechanism.
- A reproducible code repository.

Submission target. A workshop or short paper at ICIP / ICASSP 2026, or a journal extension to *IEEE Trans. Image Processing* or *IEEE Trans. Computational Imaging*; we will make a final venue decision after the W6 internal review based on experimental strength.

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